



ELSTE Data science

Some concept of Python coding and statistical inference

Sébastien Biass 

sebastien.biasse@unige.ch

Earth Sciences

Stéphane Guerrier 

Stephane.Guerrier@unige.ch

Earth Sciences

Pharmaceutical Sciences

October 15, 2025

Get ready

- Go to <https://elste-master.github.io/Data-Science/>
 - **Download slides:** Class 1 > Overview
 - **Setup VS Code:** Appendices > Setup VS Code

ELSTE Data
Science Course



ELSTE Data Science
Master Course

Class 1: Pandas >

Class 2: Descriptive stats >

Class 3: Analysis of variance >

Appendices ▾

Setup VS Code

Markdown for Jupyter

Appendices > Setup VS Code

Setup VS Code

This tutorial guides you through the installation and configuration of Visual Studio Code (VS Code) for Python development in the 0013 computer room at the University of Geneva.

Open VS Code

1. Create a folder on your **H:/** drive named **data_science**
2. Open **VS Code**

On this page

Setup VS Code

Open VS Code

Create a new Jupyter Notebook

Install the Python extensions

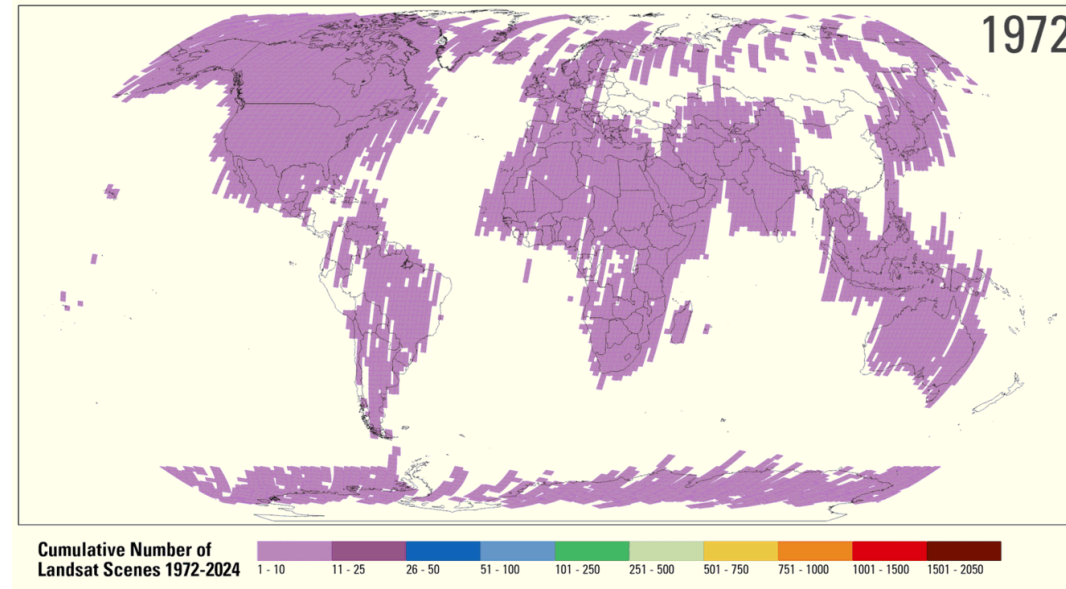
Install Jupyter

Select the right Python

Open the course material in VS Code

Why this class?

Data deluge: The big EO data landscape

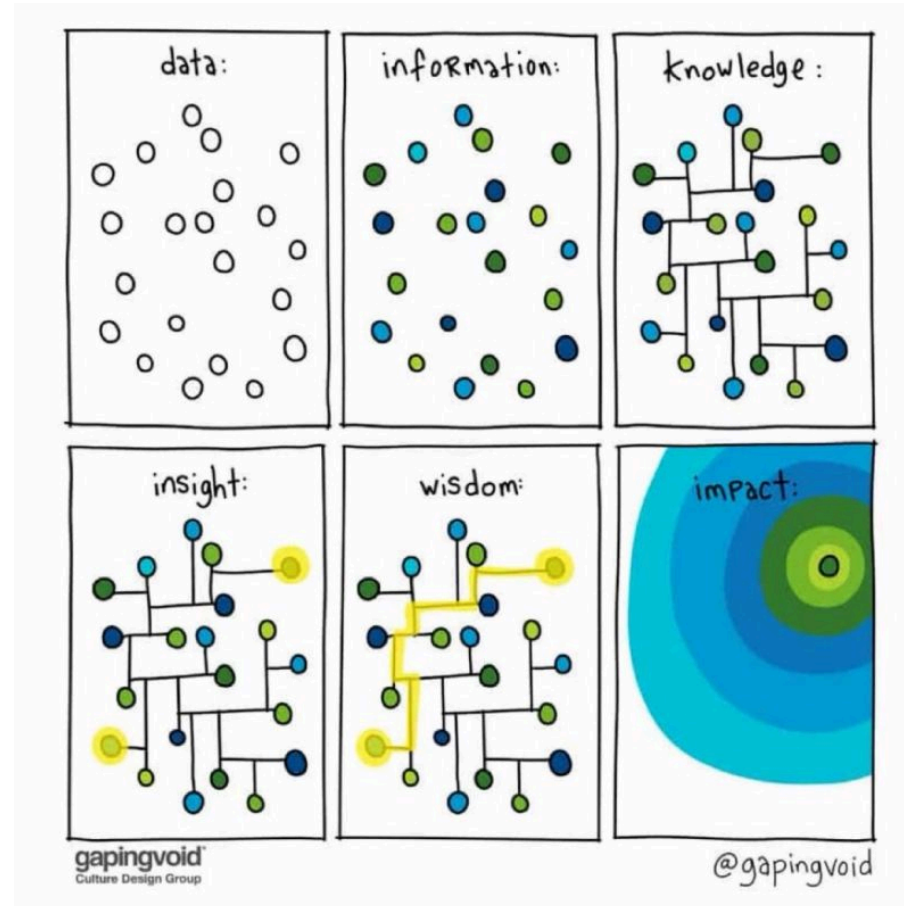


- **Landsat:**
 - Download: 53 images/day (2001) - 220'000 images/day (2017)
 - 5 million images of the Earth surface → > 5 PB
- **ESA Sentinel 1 & 2** → 4.6TB /daily!

Why this class?

As scientists, we are exposed to:

- Increasing computational power/facilities
- Increasing amount of data



What is data science

Wikipedia

Data science is an interdisciplinary academic field that uses statistics, scientific computing, scientific methods, processing, scientific visualization, algorithms and systems to extract or extrapolate knowledge from [...] data.

Class schedule

- **Fall part at UNIGE:** 6 × 3-h long sessions
- **Assistants:** Simon Thivet, Filippo Salmaso, Lionel Voirol
- **Format:** Theory, live coding and exercises

October 15 and 22

- **Seb** → Intro to data science libraries for Python → **Pandas**, **seaborn**

October 29 and November 5

- **Stéphane** → Intro to statistical method for data inference

November 12 and 19

- Flexible as a function of how we progress
- Exam format to be defined



Class objectives

- Scientific coding and stats often appear daunting
- True, there can be a steep learning curve

Get involved!!!

We want this class to be useful for your research!

Try to contextualise the course material to **your research**:

- Do you already have your own datasets you could bring to class?
- Can you use the course to formulate new research questions?
- Do you know any open-access datasets relevant to your fields?
 - If not, can you find some for the next weeks?



Why coding?

Motivations

- **Example:** A common - but unnecessarily complicated - workflow of many specialised softwares



Motivations

Motivation 1: Automation

Cotopaxi volcano

- Reconstructing eruption source parameters (ESP) from tephra deposits
 - *How do different measurement methods influence ESP estimates?*

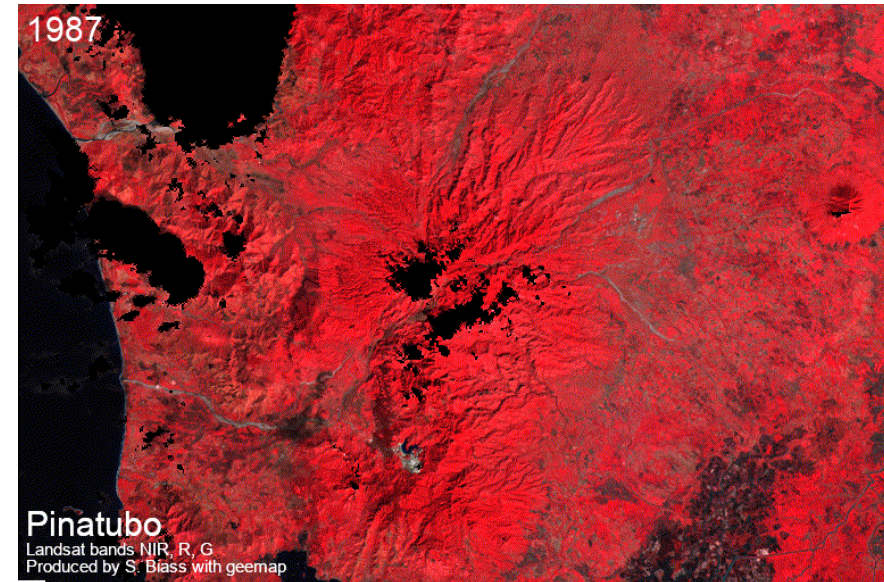


Motivations

Motivation 2: Data analysis

Example 1: Exploit catalogues of big Earth Observation data

- **Revisit big EO catalogues to infer new knowledge**
 - *What controls the impact and recovery of vegetation following eruptions?*

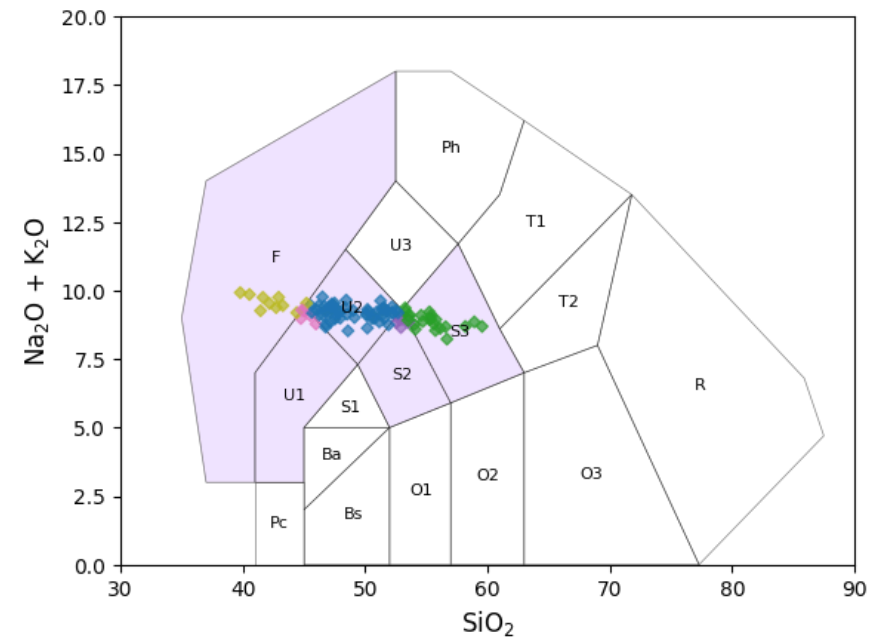


Motivations

Motivation 2: Data analysis

Example 2: Streamline global geochemical analyses

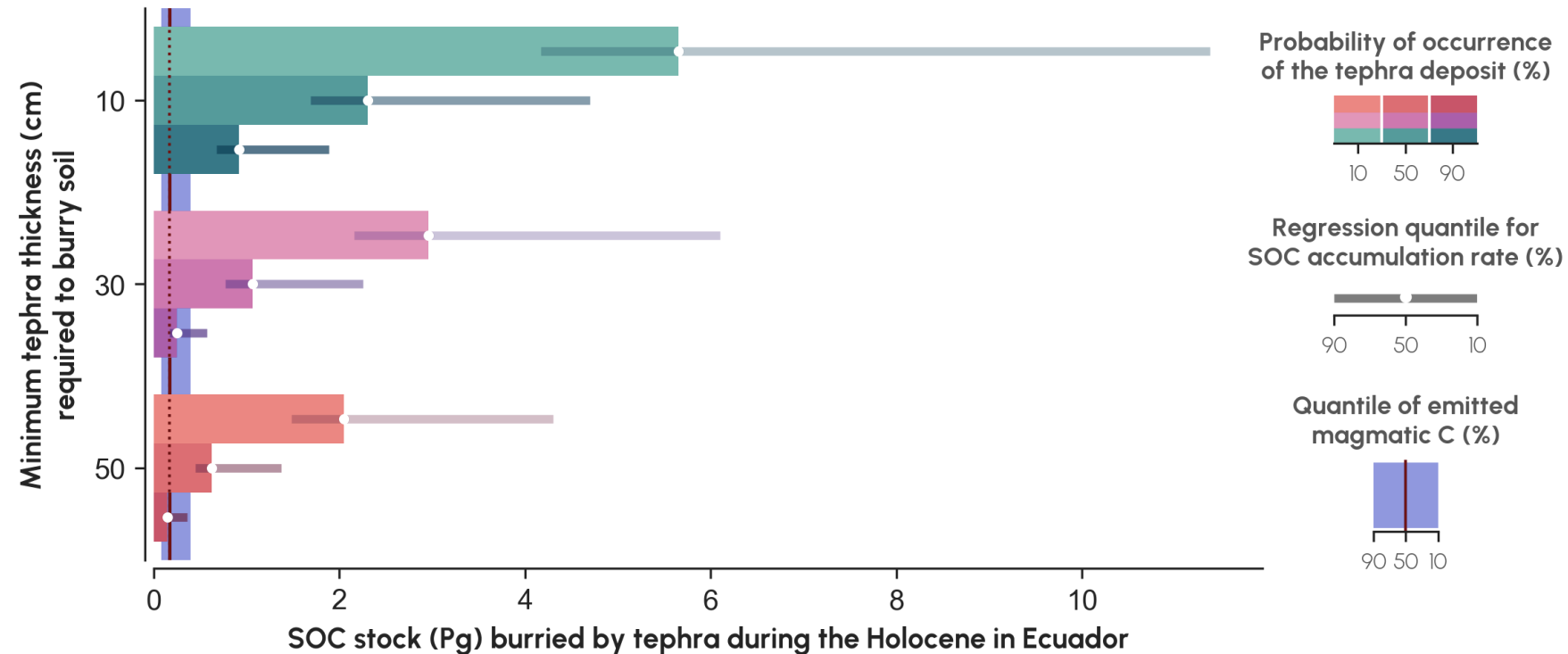
- **Access global databases**
 - e.g., *Georock* database
- **Automatic dedicated analyses/plots**
 - e.g., *TAS* diagrams and classification using **pyrolite**



Motivations

Motivation 3: Visualisation

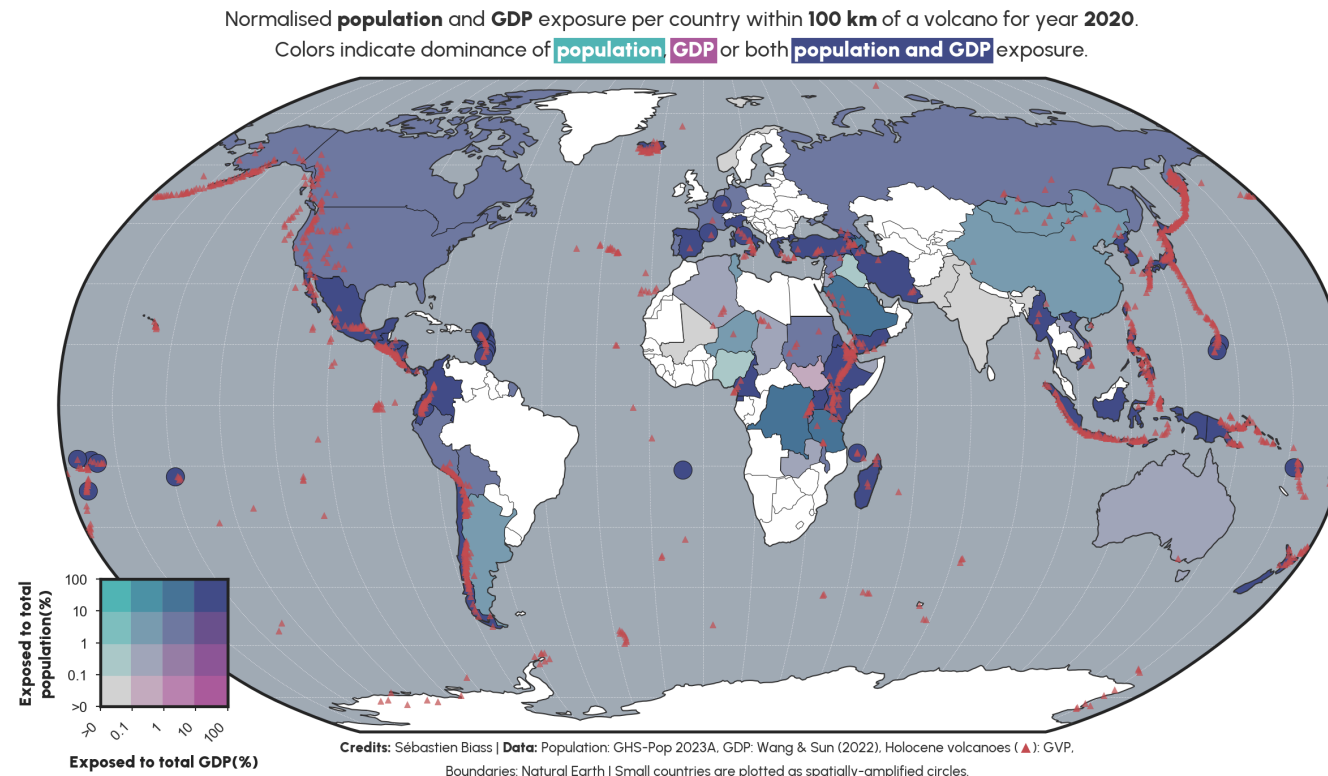
Example 1: In volcanically-active regions, soil burial after explosive eruptions capture more carbon that they emit.



Motivations

Motivation 3: Visualisation

Example 2: In Small Island States, $\geq 90\%$ of population and GDP is exposed to volcanic hazards.





Why Python?



Python's advantages ¹

1. Free

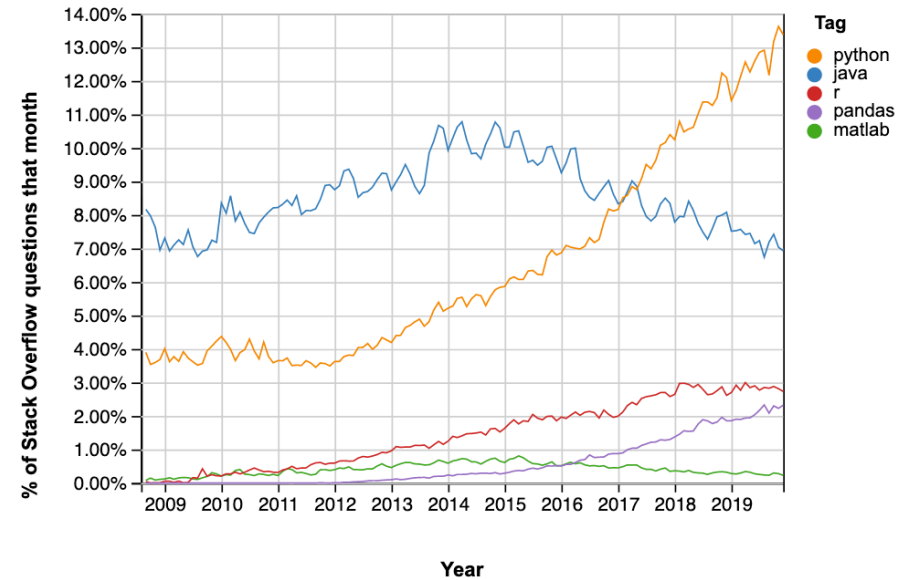
2. Open source

- Not dependent on any company
- Large online community

3. Old → stable (started in 1991)

4. **Very** popular in science

5. Relatively easy to learn



1. Inspired by the [School of Oceanography at University of Washington](#)

What is Python

High-level computing language

- Compiled → doesn't directly speak to the computer, but is *interpreted* by another language
 - Usually slower, but much, much easier to use!

An example of packages, modules and functions

In the code below, we first load the necessary packages and libraries:

- `matplotlib` is the main **visualisation** package used in Python
- `pyplot` is a module of `matplotlib` that provides easy-to-use functions for **plotting** data
- `figure` is a function of `pyplot` and is the main function to prepare a plot

```
1 # Import the packages and modules
2 from matplotlib import pyplot as plt
3 # Set up a figure for plotting
4 plt.figure(...)
```

How to use Python

Option 1: Your own computer

- Requires some setup
- No internet connection required
- Slow if computer is slow
- All data stored on your own computer

! Environment manager!

Make sure to install an **environment manager** → e.g. [Miniconda](#)

How to run Python

3 main ways to run Python...

1. Run Python `.py` scripts from the command line → *deprecated*
2. Run blocks of code from within a Python `.py` script using `ipykernel` → **similar behaviour to R/Matlab** = ❤️
3. Use **Jupyter Notebooks** in `.ipynb` files

Start coding!

- Go to <https://elste-master.github.io/Data-Science/>
→ **Setup VS Code:** Appendices > Setup VS Code

ELSTE Data
Science Course

ELSTE Data Science
Master Course

Class 1: Pandas >

Class 2:
Descriptive stats >

Class 3: Analysis of
variance >

Appendices ▾

Setup VS Code

Markdown for Jupyter

Appendices > Setup VS Code

Setup VS Code

This tutorial guides you through the installation and configuration of Visual Studio Code (VS Code) for Python development in the 0013 computer room at the University of Geneva.

Open VS Code

1. Create a folder on your **H:/** drive named **data_science**
2. Open **VS Code**

On this page

Setup VS Code

Open VS Code

Create a new Jupyter
Notebook

Install the Python
extensions

Install Jupyter

Select the right Python

Open the course material
in VS Code